

Secure Code Assessment Report

Project Title

Secure Code Assessment of Python Login System

Programming Language

Python

Database

SQLite

Objective

To analyze the source code of a Python login application, identify common security vulnerabilities, and recommend secure coding practices.

Project Description

The application is a simple login system developed in Python using SQLite. It authenticates users by verifying the entered username and password.

Vulnerabilities Identified

1. SQL Injection – User input is directly inserted into the SQL query.
2. Plain Text Password Storage – Passwords are stored without hashing.
3. Hardcoded Credentials – Fixed credentials are inserted into the database.
4. No Input Validation – User input is not validated before processing.

Recommended Fixes

- Use parameterized SQL queries.
- Store hashed passwords.
- Remove hardcoded credentials.
- Validate and sanitize user input.
- Limit login attempts.

Tools Used

Visual Studio Code, Python 3, SQLite

Conclusion

The application works correctly but contains multiple security weaknesses. Applying secure coding practices will significantly improve its security.